

Logarithmic One-at-a-Time Sensitivity Measures for the Slow Sand Filtration Model

Follow-up note

Since you vary one parameter at a time by factors 2 and 1/2, the most appropriate measure is a central sensitivity with respect to the logarithm of the parameter.

Let $L_0(t)$ denote the standard simulation and let

$$L_i^+(t) = L(t; 2\theta_i), \quad L_i^-(t) = L(t; \theta_i/2).$$

Define the time-dependent sensitivity function

$$s_i(t) = \frac{L_i^+(t) - L_i^-(t)}{2 \ln 2}.$$

This approximates

$$s_i(t) \approx \frac{\partial L(t)}{\partial \ln \theta_i} = \theta_i \frac{\partial L(t)}{\partial \theta_i}.$$

It is therefore directly comparable between parameters with different units and magnitudes.

A scalar ranking: RMS sensitivity

To obtain one scalar ranking, I recommend the root-mean-square sensitivity over the post-disturbance interval $[t_d, T]$:

$$I_i = \left[\frac{1}{T - t_d} \int_{t_d}^T s_i(t)^2 dt \right]^{1/2}$$

or, for discrete output times t_n ,

$$I_i = \left[\frac{1}{N} \sum_{n=1}^N s_i(t_n)^2 \right]^{1/2}.$$

The parameters are ranked according to decreasing I_i .

Because all parameters are perturbed by the same multiplicative factors, the denominator $2 \ln 2$ is identical for every parameter. Thus, for ranking alone, one could equivalently use

$$\tilde{I}_i = \left[\frac{1}{T - t_d} \int_{t_d}^T (L_i^+(t) - L_i^-(t))^2 dt \right]^{1/2}.$$

The logarithmic-sensitivity form is nevertheless preferable because it has a clear interpretation.

Why RMS is suitable

Using only the largest difference,

$$\max_t |L_i^\pm(t) - L_0(t)|,$$

can overemphasize a brief spike. Using the signed time average,

$$\int s_i(t) dt,$$

can underestimate influence because positive and negative effects may cancel. The RMS measure:

- accounts for the whole transient;
- does not allow cancellation;
- gives more weight to large deviations;
- retains the natural unit of L , namely log-removal units per relative parameter change.

Additional measures worth reporting

The RMS value should be the primary ranking measure, but two additional quantities would make the interpretation much more informative.

Maximum instantaneous influence

$$I_i^{\max} = \max_{t \in [t_d, T]} |s_i(t)|.$$

This identifies parameters that cause a large but short-lived deterioration or improvement.

Influence on the worst removal performance

Let

$$L_{\min,0} = \min_{t \in [t_d, T]} L_0(t),$$

and similarly define $L_{\min,i}^+$ and $L_{\min,i}^-$. Then define

$$I_i^{\min} = \frac{L_{\min,i}^+ - L_{\min,i}^-}{2 \ln 2}.$$

For safety or regulatory purposes, it may be better to report

$$D_i^{\min} = \max \left\{ \left| L_{\min,i}^+ - L_{\min,0} \right|, \left| L_{\min,i}^- - L_{\min,0} \right| \right\}.$$

A parameter may have a moderate RMS effect but a strong effect on the minimum value of L .

Check for asymmetry and nonlinearity

The two perturbations should also be compared separately:

$$\Delta_i^+(t) = L_i^+(t) - L_0(t), \quad \Delta_i^-(t) = L_i^-(t) - L_0(t).$$

For a locally linear dependence on $\ln \theta_i$, one would expect approximately

$$\Delta_i^+(t) \approx -\Delta_i^-(t).$$

A useful scalar asymmetry indicator is

$$A_i = \frac{\left[\int_{t_d}^T (L_i^+(t) + L_i^-(t) - 2L_0(t))^2 dt \right]^{1/2}}{\left[\int_{t_d}^T (L_i^+(t) - L_i^-(t))^2 dt \right]^{1/2}}.$$

A small A_i indicates approximately symmetric behavior. A large A_i means that doubling and halving the parameter do not produce opposite effects, indicating nonlinearity, saturation, or a threshold.

Recommended presentation

For every parameter, report:

$$\text{parameter} \mid I_i \mid I_i^{\max} \mid D_i^{\min}$$

and plot

$$L_i^+(t) - L_0(t), \quad L_i^-(t) - L_0(t).$$

I would use I_i , the RMS logarithmic sensitivity, for the main ranking. The measure $D_i^{\min}$ should be used as a complementary ranking when the principal concern is the lowest pathogen-removal performance after the disturbance.

For the first study, all perturbed simulations should preferably begin from the same mature-filter state. This isolates how each parameter affects the response to the disturbance. Recomputing a separate mature initial state for every perturbed parameter would instead combine sensitivity of the mature state with sensitivity of the transient response.